Paper review response plan

09/24/2026 22:11:20
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Thinking ● Reviewed repository results and validated threshold claims …

Thinking

- **Reviewed repository results and validated threshold claims**
- **Searched 9 websites**
- **Fetched repository documentation, commits, notebooks, dependencies, and feature matrices for threshold analysis**
- **Downloading CSV feature matrix locally**
- **Planning the revision**

  I'll frame the response around adding subject-level AUC confidence intervals, clarifying evaluable subjects, and supplying reproducible code. I'll avoid inventing exact results until the underlying subject-level scores are available.
- **Planned the revision**
- **Downloaded CSV feature matrix locally**
- **Calculating subject-level AUC confidence intervals**
- **Comparing evaluation protocols**

  I'm distinguishing confidence intervals from direct significance testing. Because the protocols use different labels and cohorts, I'll recommend reporting intervals while avoiding an invalid apples-to-apples comparison between their AUCs.
- **Compared evaluation protocols**
- **Calculated subject-level AUC confidence intervals**
- **Clarifying validation claims**

  I'm separating ECG-pipeline validation from classifier external validation, since the current PIPP analysis doesn't establish independent model validation. I'll also address interpretability, computational reporting, and updated methodological references where appropriate.
- **Preparing the revisions**

  I'll produce a review-response letter, revised manuscript passages, and reproducible analysis code. I'll also distinguish claims that require rerunning experiments from wording changes that can be made directly.
- **Clarified validation claims and prepared revisions**

- **Searched 9 websites**

✓ Worked for 1m 18s

Yes bro. I went through **your submitted paper**, the three reviews, and the actual LOSO notebook in your GitHub repo. This is much better than working only from the reviewer comments because there are a couple of important details in the implementation that affect how we should answer Reviewer 1.

Your paper currently reports **53 evaluable subjects / AUC 0.849 ± 0.285 / F1 0.475 ± 0.450** for the combined-label protocol and **28 evaluable subjects / AUC 0.929 ± 0.175 / F1 0.452 ± 0.438** for the PIPP-confirmed protocol. `Neonata_pain_conference_paper_submit (1).pdf`

I also checked the repository notebook: your LOSO code really does calculate AUC from `predict_proba`, while F1 comes from `model.predict()`, i.e. the default classification threshold. The notebook also confirms that your ROC figure uses **pooled AUCs of 0.806 and 0.810**, whereas the paper table reports the **mean of subject-level AUCs** (0.849 and 0.929). That's an important distinction we should make clear rather than hiding it.

Your GitHub research repository

First: I would NOT blindly follow Claude's suggested response

There is one important correction.

Claude suggested:

> *"Optimising the threshold per fold using Youden's J statistic ... is expected to substantially improve F1"*

Don't put that sentence in the paper.

You haven't actually performed that experiment, and your current code uses the default `predict()` threshold. We shouldn't claim that F1 *will* improve without measuring it.

Also, saying your probabilities are "well calibrated" is not justified by the current experiments. Your XGBoost uses `scale_pos_weight`, but you haven't reported calibration analysis.

What we *can* say accurately is:

- AUC evaluates ranking/discrimination over thresholds.
- F1 is calculated after converting probabilities into class predictions at a particular threshold.
- Your current implementation uses the model's default 0.5 decision threshold.
- Therefore, a relatively high AUC can coexist with a lower F1.
- Threshold optimization can be investigated, but if we add optimized F1, it must be done **without test-set leakage**.

Scikit-learn itself notes that the default 0.5 threshold is not necessarily optimal and that threshold tuning should use internal cross-validation rather than tuning on the same data used for fitting. (Scikit-learn)

Our plan for the three reviewers

I'd handle them like this:

Reviewer	What we should actually do
Reviewer 1	Add 95% CI + clarify AUC/F1 + strengthen limitation of n=28
Reviewer 3	Add explicit external-validation, SHAP, computational-efficiency discussion
Reviewer 4	Add 3-5 recent 2024-2026 references and integrate them into Introduction/Related Work/Discussion

And importantly, **we don't need to redesign your whole research.**

Reviewer 1 - this is the most important one

They said:

> *AUCs show substantial variability, yet no confidence intervals or statistical comparisons support the claims...*

There are actually **three separate things** here.

---

1. Add 95% confidence intervals

Your current table is:

Evaluation Group	Subjects	Mean AUC	Mean F1
Broad	53	0.849 ± 0.285	0.475 ± 0.450
PIPP	28	0.929 ± 0.175	0.452 ± 0.438

`Neonata_pain_conference_paper_submit (1).pdf`

We should change this to something like:

Evaluation Group	Subjects	Mean AUC (95% CI)	Mean F1
Broad condition-based	53	**0.849 (0.770-0.928)**	0.475 ± 0.450
PIPP-confirmed	28	**0.929 (0.860-0.998)**	0.452 ± 0.438

Those intervals are the **approximate 95% t-based CIs derived from the mean, SD and n already reported in your paper**.

For the actual revision, however, I recommend **rerunning from your stored per-subject AUC values and calculating the CI directly**, rather than relying on the rounded table values.

Add this to your notebook

Modify your LOSO loop so that you save the subject-level results:

```
from scipy import stats
import numpy as np
import pandas as pd

all_auc = []
all_f1 = []
all_subjects = []
```

```
for subj in unique_subj:

    test_mask  = subjects == subj
    train_mask = \~test_mask

    X_train, y_train = X[train_mask], y[train_mask]
    X_test, y_test   = X[test_mask], y[test_mask]

    # Skip single-class test subjects
    if len(np.unique(y_test)) < 2:
        continue

    scaler = StandardScaler()

    X_train = scaler.fit_transform(X_train)
    X_test  = scaler.transform(X_test)

    scale_pw = (y_train == 0).sum() / max((y_train == 1).sum(), 1)

    model = xgb.XGBClassifier(
        n_estimators=300,
        max_depth=3,
        learning_rate=0.03,
        min_child_weight=3,
        subsample=0.8,
        colsample_bytree=0.8,
        scale_pos_weight=scale_pw,
        eval_metric="logloss",
        random_state=42,
        verbosity=0
    )

    model.fit(X_train, y_train)

    y_prob = model.predict_proba(X_test)[:, 1]
    y_pred = model.predict(X_test)

    auc = roc_auc_score(y_test, y_prob)
    f1 = f1_score(y_test, y_pred, zero_division=0)

    all_auc.append(auc)
```

```
    all_f1.append(f1)
    all_subjects.append(subj)


# Convert to arrays
all_auc = np.array(all_auc)
all_f1 = np.array(all_f1)


# Mean and standard deviation
mean_auc = np.mean(all_auc)
std_auc = np.std(all_auc, ddof=1)

mean_f1 = np.mean(all_f1)
std_f1 = np.std(all_f1, ddof=1)


# 95% CI for mean AUC
sem_auc = stats.sem(all_auc)

ci_low, ci_high = stats.t.interval(
    0.95,
    df=len(all_auc) - 1,
    loc=mean_auc,
    scale=sem_auc
)


print("=" * 60)
print("LOSO RESULTS")
print("=" * 60)

print(f"Subjects evaluated : {len(all_auc)}")

print(
    f"Mean AUC : {mean_auc:.3f} ± {std_auc:.3f}"
)

print(
    f"95% CI   : [{ci_low:.3f}, {ci_high:.3f}]"
)
```

```
print(
    f"Mean F1  : {mean_f1:.3f} ± {std_f1:.3f}"
)


# Save subject-level results
results = pd.DataFrame({
    "subject": all_subjects,
    "auc": all_auc,
    "f1": all_f1
})

results.to_csv(
    "loso_subject_level_results.csv",
    index=False
)

print("\nSaved subject-level results.")
```

Why I prefer this

Your paper currently says that the large variation is expected because each LOSO test fold contains only a small number of recordings.
`Neonata_pain_conference_paper_submit (1).pdf`

The CI gives the reviewer an actual quantitative uncertainty interval rather than only:

`0.849 ± 0.285`

---

2. Don't perform a fake "statistical comparison"

This part is subtle.

The reviewer asks for:

> *"confidence intervals or statistical comparisons"*

It does **not** mean we must manufacture a p-value.

Your two numbers:

- 0.849 = broad condition-based evaluation
- 0.929 = PIPP-confirmed evaluation

are **not perfectly equivalent experiments**.

The paper itself says the PIPP protocol is a clinically validated subset with a different labeling strategy. `Neonata_pain_conference_paper_submit (1).pdf`

Therefore, I would explicitly say:

> *We added 95% confidence intervals. Because the two evaluation protocols use different label definitions and have different evaluable subject populations, we did not perform a direct statistical significance test between the two mean AUCs, as such a comparison would not represent an equivalent experimental condition.*

That's scientifically defensible.

And importantly, we should stop saying things like:

> *"PIPP is statistically significantly better."*

We're not going to claim that.

---

3. Reviewer 1's AUC/F1 concern

This is actually something we can explain very nicely because I checked your code.

Your notebook does:

```
y_prob = model.predict_proba(X_test)[:, 1]

auc = roc_auc_score(y_test, y_prob)

y_pred = model.predict(X_test)

f1 = f1_score(y_test, y_pred)
```

So:

AUC

uses the continuous prediction:

```
0.13
0.27
0.48
0.61
0.73
0.91
...
```

and evaluates discrimination across thresholds.

Whereas F1 uses the final class prediction:

```
probability >= 0.5 → pain
probability < 0.5  → non-pain
```

The default threshold for probability-based binary classification is 0.5, while threshold tuning can change class predictions without changing ROC-AUC. (Scikit-learn)

So **this is the paragraph I recommend putting in your Discussion.**

Reviewer 1 - AUC/F1 Discussion Revision

The apparent discrepancy between the relatively high AUC values and the lower F1-scores is related to the different properties of the two metrics and the decision threshold used for classification. AUC evaluates the model's ability to rank positive and negative samples across a range of decision thresholds, whereas F1-score is calculated from the discrete class predictions obtained at a specific threshold. In the present implementation, F1-score was computed using the XGBoost classifier's default probability decision threshold of 0.5. Therefore, a model may exhibit good ranking discrimination while producing a lower F1-score at the selected operating threshold, particularly in the presence of class imbalance and small subject-level test sets. Threshold optimization was not performed in the present study because the primary evaluation objective was subject-independent discrimination using LOSO cross-validation. Future work will investigate leakage-free threshold optimization and calibration within the LOSO framework to determine whether an operating point better suited to clinical decision-making can improve classification metrics such as F1-score.

This is much safer than Claude's version.

It doesn't claim an improvement that you haven't measured.

One more thing I want to fix in your wording

Your current paper says:

> *"Both curves remain well above the random classification baseline"*

but the ROC curves are based on pooled AUCs of **0.806 and 0.810**, while your headline numbers are the **mean per-subject AUCs** of 0.849 and 0.929.

Your notebook explicitly shows this:

```
Mean AUC (per-subj): 0.849
Pooled AUC         : 0.806

Mean AUC (per-subj): 0.929
Pooled AUC         : 0.810
```

This distinction is important.

I recommend adding this sentence to Methods/Results:

> *"For the primary LOSO analysis, AUC was summarized as the arithmetic mean of the subject-level AUCs. Pooled out-of-fold predictions were additionally used to construct the aggregate ROC curves shown in Fig. 5; consequently, the pooled AUC values differ from the mean subject-level AUCs reported in Table IV."*

That will prevent a reviewer from wondering why the ROC curve doesn't correspond to 0.929.

Reviewer 1 - small PIPP cohort

The reviewer specifically says:

> *"The small PIPP cohort (n=28) also limits LOSO generalisability."*

We should **agree with them**, not fight it.

Your current limitation says the evaluable subject count is constrained by PIPP availability and the requirement for mixed-class test sets. `Neonata_pain_conference_paper_submit (1).pdf`

I'd replace that section with:

Revised Limitations

The principal limitation of this study is the limited number of evaluable subjects under the strict PIPP-confirmed LOSO protocol. Although PIPP scores were available for a larger subset of the UCL cohort, only 28 subjects satisfied the criteria required for subject-level evaluation with both pain and non-pain classes present in the held-out test set. Consequently, the PIPP-confirmed results should be interpreted as preliminary evidence of subject-independent discrimination rather than definitive evidence of generalisability to the broader neonatal population. Larger multi-centre cohorts with standardized clinical pain annotations are required to establish the robustness and clinical transportability of the proposed framework.

That's much stronger academically.

Reviewer 3

Reviewer 3 basically said:

> *methodology is technically sound and results promising*

and asked for:

1. external validation
2. interpretability
3. computational analysis

You already have some of this in Future Work. `Neonata_pain_conference_paper_submit (1).pdf`

But we'll make it explicit.

Important distinction

Your PICS dataset **is not external validation of the pain classifier**.

Your paper says PICS was used to validate ECG preprocessing and HRV extraction, while UCL was used for the actual classifier. `Neonata_pain_conference_paper_submit (1).pdf`

So don't tell Reviewer 3:

> *"We performed external validation using PICS."*

That would be incorrect.

Instead:

> *"PICS provides independent validation of the ECG preprocessing pipeline, whereas external validation of the complete multimodal classifier remains future work."*

That's a very good response.

Revised Future Work

Revised Future Work

Future work will extend the present framework in three directions. First, the complete multimodal classifier will be externally validated on an independent neonatal cohort to assess transportability across institutions, acquisition settings, and patient populations. The present PICS experiment provides independent validation of the ECG preprocessing and HRV extraction pipeline, but does not constitute external validation of the complete pain classifier. Second, explainable AI methods, particularly SHAP-based feature attribution, will be investigated to quantify the contribution of EEG, ECG, and respiration features to individual predictions and to identify physiologically meaningful biomarkers. Third, computational profiling will be performed by measuring training and inference time, memory requirements, and model complexity to assess the feasibility of eventual real-time or edge-based bedside deployment.

This directly answers all three things Reviewer 3 asked for.

SHAP-based interpretation is also a well-established way of explaining XGBoost predictions in clinical ML studies. (Springer)

Reviewer 4 - recent citations

This one is easy to fix, and actually your paper is currently a little weak here.

Your current references include Hamwi 2025 and Roué 2025, which are good, but we can add several **2024-2026** works.

Your paper already discusses Hamwi and Roué. `Neonata_pain_conference_paper_submit (1).pdf`

I recommend adding at least these:

New Reference A - 2025

Hamwi et al. is already there, but I'd keep it and make sure the citation is correctly formatted.

It is a 2025 *PAIN Reports* paper using ML for EEG-based neonatal pain-related vs non-pain-related stimulus classification. (DOI)

New Reference B - 2025

Roué et al. is already there.

Their study used EEG and ML and reported an AUC around 0.90 with gradient boosting, which is directly relevant to your comparison. (PubMed Central (PMC))

New Reference C - 2024

Add Shafiee & Daliri:

> *R. Shafiee and M. R. Daliri, "Decoding of pain during heel lancing in human neonates with EEG signal and machine learning approach," Scientific Reports, vol. 14, Art. no. 31244, 2024, doi: 10.1038/s41598-024-82631-0.*

This is directly relevant because it uses neonatal heel-lance EEG and ML. (Nature)

New Reference D - 2024

Add the multimodal dataset paper:

> *N. Yang et al., “Developing and Validating a Multimodal Dataset for Neonatal Pain Assessment to Improve AI Algorithms With Clinical Data,” Advances in Neonatal Care, vol. 24, no. 6, pp. 578-585, 2024, doi: 10.1097/ANC.0000000000001205.*

This is particularly useful for your discussion of **ground-truth quality and multimodal neonatal pain assessment**. (PubMed)

New Reference E - 2026

This is the one I particularly recommend because it is **very recent and directly aligned with Reviewer 4's request**.

> *Q. Zhou et al., “Application of Intelligent Neonatal Pain Assessment Tools Based on Multimodal Data Fusion: A Systematic Review,” Journal of Pain Research, vol. 19, 2026, Art. no. 578721, doi: 10.2147/JPR.S578721.*

The review searched studies through September 2025 and specifically examines multimodal intelligent neonatal pain assessment. Importantly, it reports that only **3 of 9 included studies used external validation**, which supports your decision to explicitly identify external validation as future work rather than pretending your current study already has it. (Taylor & Francis Online)

Where to put these citations

Don't just dump five references at the end.

Use them strategically.

Introduction

Your current:

> *"Recently, several studies have leveraged the publicly available UCL neonatal dataset..."*

can become:

Revised Recent-Work Paragraph

Recent studies have increasingly applied machine learning to neonatal physiological recordings for automated pain assessment. Hamwi et al. [5] evaluated multiple machine learning models for distinguishing pain-related from non-pain-related EEG activity in preterm infants, while Roué et al. [6] investigated gradient-boosting approaches for objective detection of acute procedural pain using EEG. Other recent work has explored EEG-based pain decoding using time-frequency features and conventional machine learning classifiers [16]. In parallel, multimodal neonatal pain datasets and clinically validated labeling strategies have received increasing attention, emphasizing the importance of reliable ground-truth annotation for AI development [17]. A recent systematic review of intelligent multimodal neonatal pain assessment further highlighted the promise of multimodal approaches while identifying limited use of external validation across existing studies [18].

Then your **research gap** becomes much more defensible.

Add this to Discussion

Your paper currently says:

> *"Labels which are Clinically validated reduce ambiguity and improve the discriminative ability of the classifier."*

That's a little too strong because you're interpreting your own result causally.

I'd soften it:

Revised Discussion Paragraph

The increase in mean subject-level AUC from 0.849 under the broad condition-based protocol to 0.929 under the PIPP-confirmed protocol suggests that the choice of label definition substantially affects observed classification performance. The PIPP-confirmed protocol provides a more clinically anchored definition of the positive class, whereas the broader condition-based protocol relies on the experimental procedure as the primary label source. However, the

two protocols differ in both cohort composition and label definition, and therefore the difference in AUC should be interpreted descriptively rather than as evidence of statistically superior performance of one labeling strategy. This observation is consistent with recent work emphasizing the importance of reliable clinical annotation and standardized multimodal datasets for neonatal pain AI research [17], as well as recent reviews highlighting heterogeneity in outcome definitions across multimodal neonatal pain studies [18].

This is much more scientifically careful.

I would also revise your Conclusion

Your current conclusion says:

> *"These results confirm that clinically validated labels and multimodal physiological integration jointly improve subject-independent pain classification."*

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"confirm" and "improve" are too strong given the reviewer concern and the different cohorts.

Change it to:

Revised Conclusion

This study presented a multimodal machine learning framework for neonatal pain classification using EEG, ECG, and respiration signals with subject-independent LOSO evaluation. The ECG preprocessing and HRV extraction pipeline was independently assessed using the PICS dataset, while multimodal pain classification was evaluated on the UCL neonatal pain dataset. The framework achieved mean subject-level AUCs of 0.849 for the broad condition-based protocol and 0.929 for the PIPP-confirmed protocol. The relatively wide variability across LOSO folds highlights the uncertainty associated with small subject-level test sets and the limited size of the PIPP-confirmed evaluable cohort. These findings provide preliminary evidence that combining cortical and autonomic physiological features can support subject-independent neonatal pain classification, while

> larger independent cohorts, external validation, explainability analysis, and computational profiling remain necessary before clinical translation.

That conclusion will survive Reviewer 1 much better.

One thing I would NOT add

Don't write:

> *"Our model outperforms existing methods."*

Your own Table V says:

- Hamwi: 0.94
- Roué: 0.90
- Yours: 0.929

and Hamwi's 0.94 was obtained in a specific gestational-age subgroup.
`Neonata_pain_conference_paper_submit (1).pdf`

Your current comparison is already reasonably careful. Keep it descriptive:

> *"performance comparable to recent studies"*

rather than:

> *"superior"*

Reviewer-response document

You will also need a **response-to-reviewers** document. I would submit something like this:

> **Response to Reviewers**
>
> ### Response to Reviewer #1

We sincerely thank the reviewer for highlighting the uncertainty associated with the LOSO results and the apparent discrepancy between AUC and F1-score.

Comment: The reported AUCs show substantial variability, yet no confidence intervals or statistical comparisons support the reported performance. The F1-scores are also low and highly variable.

Response: We agree that reporting only mean ± standard deviation does not fully characterize the uncertainty of subject-level performance. We have therefore added 95% confidence intervals for the mean subject-level AUCs based on the LOSO subject-level results. The revised results table now reports the mean AUC together with its 95% confidence interval. We have also clarified that the two evaluation protocols use different label definitions and evaluable subject populations; consequently, we do not interpret the difference between the two AUCs as a formal statistical comparison between equivalent experimental conditions.

We have additionally expanded the Discussion to explain the difference between AUC and F1-score. AUC measures discrimination across decision thresholds, whereas F1-score is calculated from discrete predictions at a specific operating threshold. In our implementation, F1-score was calculated using the XGBoost classifier's default decision threshold. We now explicitly state this and identify leakage-free threshold optimization and probability calibration as future work rather than claiming an unverified improvement in F1-score.

Comment: The small PIPP cohort (n=28) limits LOSO generalisability.

Response: We agree with this limitation. The revised manuscript explicitly states that only 28 subjects satisfied the criteria for evaluable subject-level LOSO testing under the PIPP-confirmed protocol. We now emphasize that these results should be considered preliminary evidence of subject-independent discrimination rather than definitive evidence of generalisability. We also identify larger multi-centre cohorts with standardized clinical pain annotations as an important direction for future validation.

Response to Reviewer #3

We thank the reviewer for the positive assessment of the multimodal framework and subject-level validation strategy.

Comment: Further improvements in external validation, model interpretability, and computational analysis would strengthen the contribution.

Response: We have expanded the Future Work section to address all three points explicitly. First, external validation of the complete multimodal classifier on an independent neonatal cohort is identified as a priority for assessing transportability. We clarify that the PICS dataset was used only for independent validation of the ECG preprocessing and HRV extraction pipeline and therefore does not constitute external validation of the complete pain classifier. Second, SHAP-based explainability is proposed to quantify the contribution of EEG, ECG, and respiration features to individual predictions. Third, computational profiling, including training time, inference time, memory requirements, and model complexity, is identified as necessary for evaluating eventual real-time or edge-based bedside deployment.

Response to Reviewer #4

We thank the reviewer for recommending improvements to the reference section.

Comment: Improve the citations and add recent references.

Response: We have revised the Introduction and Related Work sections and added recent literature published during 2024-2026 covering EEG-based neonatal pain classification, multimodal neonatal pain datasets, and multimodal intelligent neonatal pain assessment. In particular, recent studies by Hamwi et al. (2025), Roué et al. (2025), Shafiee and Daliri (2024), and Yang et al. (2024), together with a 2026 systematic review of multimodal intelligent neonatal pain assessment, have been incorporated to provide a more up-to-date representation of the field.

The revision sequence I recommend

Don't edit the paper randomly. Do it in this order:

Step 1 - rerun the LOSO analysis

First save:

```
subject_id
AUC
```

```
F1
```

for both protocols.

Step 2 - calculate

For both:

```
mean AUC
SD
95% CI
```

Step 3 - inspect F1

Don't optimize it yet.

First understand the distribution of:

```
probabilities
threshold = 0.5
F1
```

Then, if we decide to add threshold optimization, we'll implement it **inside the LOSO training folds**, not on the test subjects.

Step 4 - fix Table IV

Change:

```
0.849 ± 0.285
0.929 ± 0.175
```

to:

```
0.849 (95% CI: ...)
0.929 (95% CI: ...)
```

while keeping F1 as mean ± SD unless we also calculate its CI.

Step 5 - fix the AUC/F1 discussion

Use the revised paragraph above.

Step 6 - fix limitations

Explicitly state why only **28** PIPP subjects are evaluable.

Step 7 - clarify pooled vs mean AUC

This is important because your notebook actually has:

```
Mean subject AUC
        ≠
Pooled ROC AUC
```

and reviewers may notice.

Step 8 - update Future Work

External validation + SHAP + computational profiling.

Step 9 - add recent references

I'd use the 2024-2026 papers above. The 2026 systematic review is especially useful because it directly discusses the current state of multimodal neonatal pain AI and the limited use of external validation. (Taylor & Francis Online)

And bro, there is actually a positive point here

Your repository gives us something very useful for Reviewer 1: **the reviewer asked about an apparent AUC/F1 inconsistency, and we can trace exactly how those metrics were produced.**

Your implementation isn't hiding anything:

- subject-level LOSO
- `StandardScaler` fitted inside each fold
- `scale_pos_weight` calculated from training data
- XGBoost
- probability-based AUC

- default classification prediction for F1

That makes the response much more defensible.

Your paper also already explicitly describes fold-wise standardization and class-imbalance handling. `Neonata_pain_conference_paper_submit (1).pdf`

The next thing I would do is not rewrite the whole paper yet. We should first run the exact revised statistical analysis on your actual LOSO outputs, because the final CI numbers-and especially whether we should add threshold-optimized F1-should come from the experiment, not from prose.

If you upload the `ucl_feature_matrix_clean.csv` from the repo (or your latest LOSO-results CSV if you have one), I can calculate the exact 95% CIs, check the subject-level AUC/F1 distributions, and then we can make the **final Table IV + exact revised Results/Discussion + reviewer response** consistent with the actual numbers.

Sources

TunedThresholdClassifierCV — scikit-learn 1.9.0 documentation - sklearn sklearn.org

3.3. Tuning the decision threshold for class prediction — scikit-learn 1.5.2 documentation scikit-learn.org

XGBoost-SHAP-based interpretable diagnostic framework for knee osteoarthritis: a population-based retrospective cohort study | Arthritis Research & Therapy | Springer Nature Link link.springer.com

Machine learning classification of EEG responses... : PAIN Reports doi.org

Objective Detection of Newborn Infant Acute Procedural Pain Using EEG and Machine Learning Algorithms - PMC pmc.ncbi.nlm.nih.gov

Decoding of pain during heel lancing in human neonates with EEG signal and machine learning approach | Scientific Reports nature.com

Developing and Validating a Multimodal Dataset for Neonatal Pain Assessment to Improve AI Algorithms With Clinical Data - PubMed pubmed.ncbi.nlm.nih.gov

Full article: Application of Intelligent Neonatal Pain Assessment Tools Based on Multimodal Data Fusion: A Systematic Review tandfonline.com

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